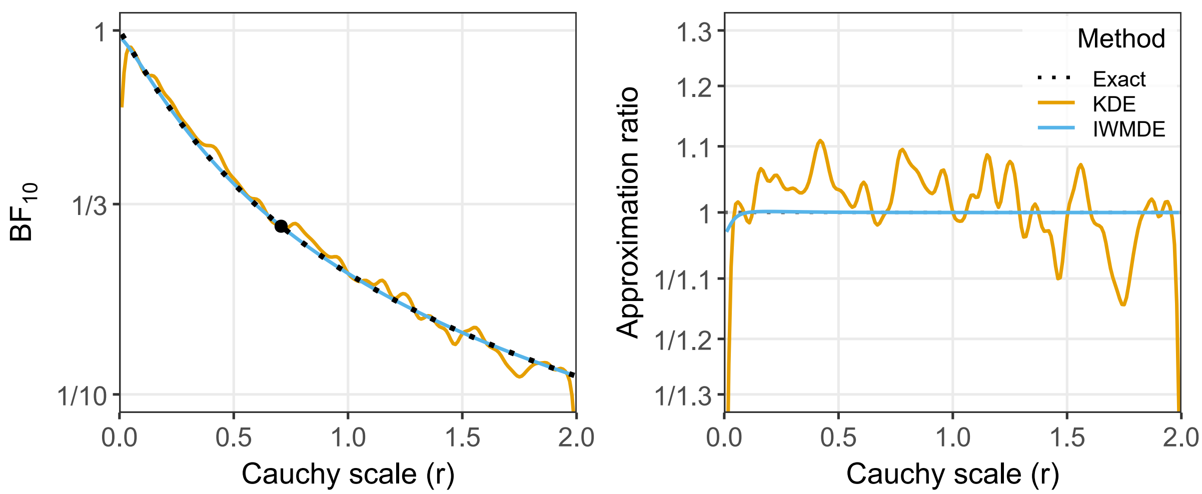

Figure 1: Univariate sensitivity analysis for the default Bayesian t-test applied to the Oosterwijk facial-feedback data. Left panel: Bayes factor BF_{10} as a function of the Cauchy scale parameter r; the black dotted line shows the exact solution, the colored lines show the kernel density estimator (KDE) and the importance-weighted marginal density estimator (IWMDE), and the filled circle marks the anchor at $r_0 = \sqrt{2}/2$. Right panel: approximation ratio $\mathrm{BF}_{\text{approx}}/\mathrm{BF}_{\text{exact}}$ for KDE and IWMDE.

($M = 4.63$, $\mathrm{SD} = 1.48$) and 57 participants in the pout condition ($M = 4.87$, $\mathrm{SD} = 1.32$). This data set was previously analyzed by Gronau et al. (2020) and Bartoš & Wagenmakers (2023). All extended models are fit in Stan (Carpenter et al., 2017); for the t-test examples, anchor Bayes factors are computed exactly via the `BayesFactor` package (Morey & Rouder, 2015).

3.1.1 Univariate Sensitivity Analysis

The default Bayesian t-test (Rouder et al., 2009) assigns a Cauchy prior on effect size, $\mathcal{H}_1 : \delta \sim \text{Cauchy}(0, r)$, where the scale r controls the width of the prior. The sensitivity parameter is $\gamma = r$ and the analysis is one-dimensional. The anchor is set to the default scale $r_0 = \sqrt{2}/2$, and the extended model places a $\text{Uniform}(0, 2)$ hyper-prior on r.

Figure 1 compares the approximated sensitivity curves to the exact solution computed by the `BayesFactor` package (Morey & Rouder, 2015). The sensitivity analysis demonstrates that we would find evidence in favor of the null hypothesis regardless of the prior width. Both KDE and IWMDE closely track the exact Bayes factor across the full range of the Cauchy scale (left panel). The IWMDE curve is nearly indistinguishable from the exact solution, with the approximation ratio remaining within a few percent of unity even at the boundaries of the support (right panel). The KDE approximation is slightly less precise at the edges of the parameter space, particularly for very small and very large values of r, though the discrepancy remains modest. For example, at $r = 0.01$ and $r = 2$, the KDE approximation ratio deviates by up to approximately 10–20%, whereas the IWMDE stays within 1–2%. Near the anchor, both methods are essentially exact, as expected from the construction of the density ratio.

3.1.2 Number of MCMC Samples

An important practical question is how many posterior samples the extended model needs to produce before the density ratio becomes sufficiently accurate. To investigate this, we refit the extended model with five different MCMC sample sizes (3,000, 10,000, 30,000, 100,000, and 300,000 total post-warmup draws) and evaluate the associated KDE and IWMDE approximations.

Figure 2 displays the approximation ratio for both methods across the five sample sizes. The difference in behavior is striking. The IWMDE (right panel) produces a nearly flat approximation ratio even with as few as 3,000 posterior samples: the curve is virtually indistinguishable from the exact solution at all sample sizes considered. In contrast, the KDE (left panel) exhibits noticeable variability across the entire range of r, and this variability diminishes only slowly with increasing sample size. Even at 300,000 samples, the KDE approximation ratio shows residual wiggle, particularly near the boundaries. This pattern is consistent with the theoretical properties of the two estimators: the IWMDE

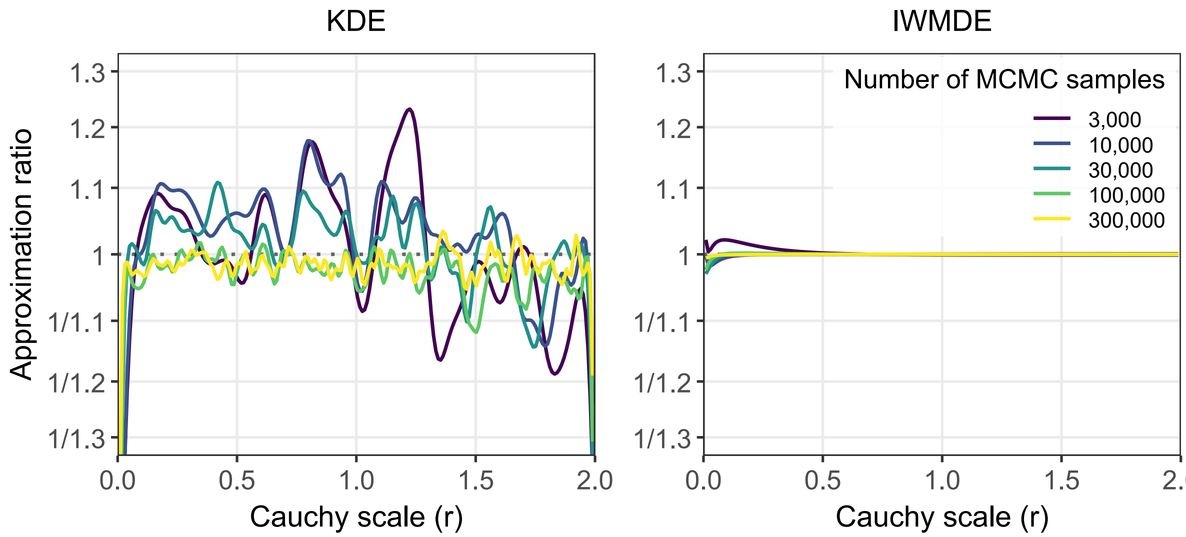

Figure 2: Effect of Markov chain Monte Carlo (MCMC) sample size on the approximation ratio for the univariate Bayesian t-test sensitivity analysis. Left panel: kernel density estimator (KDE). Right panel: importance-weighted marginal density estimator (IWMDE). Each curve corresponds to a different number of posterior draws from the extended model (3,000 to 300,000). The IWMDE achieves near-perfect accuracy even with $3{,}000$ draws, whereas the KDE exhibits visible variability that decreases slowly with sample size.

exploits the known functional form of the prior linking γ and the model parameters, effectively Rao-Blackwellizing the density estimate, whereas the KDE treats the marginal posterior of γ as an opaque distribution and must estimate its shape entirely from the empirical sample. Fitting semi-parametric or parametric density models, such as logspline, log-concave, or truncated parametric families, can improve upon the plain KDE in the univariate case (see Appendix 4 for the full set of methods considered), but these alternatives are generally more variable than the IWMDE and do not generalize well to multivariate settings.

3.1.3 Bivariate Sensitivity Analysis

The informed Bayesian t-test of Gronau et al. (2020) places a normal prior on effect size, $\mathcal{H}_1 : \delta \sim \mathrm{N}(\mu_\delta, \sigma_\delta^2)$. This provides a test case for the bivariate extension (Section 2.1), with $\boldsymbol{\gamma} = (\mu_\delta, \sigma_\delta) \in [0, 1] \times (0, 1]$ and anchor $\boldsymbol{\gamma}_0 = (0.5, 0.5)$. The extended model places independent Uniform$(0, 1)$ hyper-priors on μ_δ and σ_δ.

Figure 3 demonstrates that there is evidence in favor of the null hypothesis across the whole range of the examined parameterizations of the alternative hypothesis. Both KDE and IWMDE recover the overall structure of the exact $\log \mathrm{BF}_{10}$ surface, but the IWMDE provides a visibly smoother and more faithful approximation. Both approximations perform less well in the region where μ_δ is large and σ_δ is small. The lower right corner of the parameter space is especially challenging: the combination of a large prior mean and a narrow prior standard deviation concentrates the prior far from the observed data (which show a small negative effect), producing very small Bayes factors deep in the tail of the posterior distribution of $\boldsymbol{\gamma}$. The KDE approximation exhibits more pronounced contour irregularities in this region, reflecting the difficulty of nonparametric density estimation in areas with sparse posterior mass. The IWMDE does slightly better by exploiting the known normal-prior structure to produce smoother density ratio estimates even where posterior samples are scarce. An edge of the lower-right region (μ_δ near 1, σ_δ near 0.05) lacked sufficient posterior samples for reliable evaluation and is therefore left blank.

It is worth noting that this data set represents a particularly demanding test case for sensitivity analysis: the data provide substantial evidence in favor of the null hypothesis across the entire parameter space, pushing the Bayes factor into a range where small errors in the density ratio are amplified multiplicatively.

3.2 Bayesian Model-Averaged Meta-Analysis

The second example applies the method to Bayesian model-averaged meta-analysis, where inference is averaged across multiple component models. We re-analyze the $K = 9$ experimental studies of precognition reported by Bem (2011),

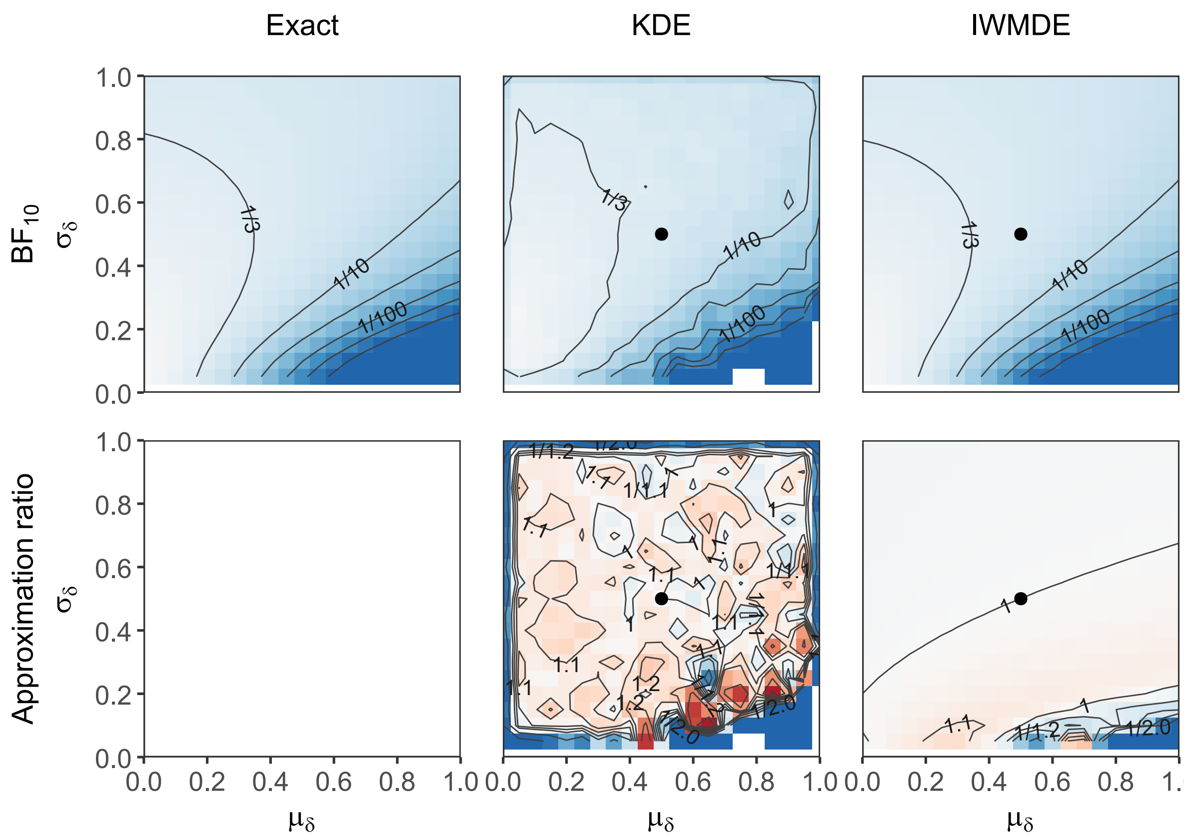

Figure 3: Bivariate sensitivity analysis for the informed Bayesian t-test. Columns correspond to the exact computation, the kernel density estimator (KDE), and the importance-weighted marginal density estimator (IWMDE). Top row: Bayes factor BF_{10} as a function of the prior mean μ_δ and prior standard deviation σ_δ; contour labels show BF_{10} levels (darker colors indicate more extreme evidence), and the filled circles mark the anchor at $(\mu_\delta, \sigma_\delta) = (0.5, 0.5)$ in the approximation panels. Bottom row: approximation ratio $\text{BF}_{\text{approx}}/\text{BF}_{\text{exact}}$ (darker colors indicate more extreme ratio, ratios larger than one in red, ratios lower than one in blue); the lower-left panel is blank because the exact computation has no approximation ratio.

using data from the `RoBMA` package (Bartoš & Maier, 2020). This data set was previously used to demonstrate the performance of publication bias adjustment methods (see Bartoš et al., 2022), as publication bias and other questionable research practices are arguably the most plausible explanation for the evidence of the effect (e.g., Francis, 2012; Schimmack, 2012; Alcock, 2011).

We assess sensitivity of the results to the specification of the alternative hypothesis on the overall effect. The sensitivity parameter is $\gamma = \sigma_\mu$, the standard deviation of the normal prior distribution on μ, $\mu \sim \text{N}(0, \sigma_\mu^2)$. The sensitivity range is specified as $[0, 2]$ with the default anchor at $\sigma_\mu = 1$. The heterogeneity prior $\tau \sim \text{Inverse-Gamma}(1, 0.15)$ is held fixed throughout.

3.2.1 Bayesian Model-Averaged Meta-Analysis

The (publication-bias-unadjusted) Bayesian model-averaged meta-analysis consists of four models formed by crossing the presence or absence of μ with fixed-effect ($\tau = 0$) or random-effects ($\tau \sim \text{Inverse-Gamma}(1, 0.15)$) structure, as described in Section 2.2. The quantity of interest is the inclusion Bayes factor $\text{BF}_{\text{incl}}(\sigma_\mu)$, which contrasts the two models that include μ against the two null models. At the anchor, the inclusion Bayes factor and the per-model marginal likelihoods are obtained via bridge sampling through the `RoBMA` package.

Both implementation strategies described in Section 2.2 are compared. The first strategy (bridge encompassing; top row of Figure 4) fits two separate extended JAGS models (Plummer, 2003), one for the fixed-effect alternative and one for